

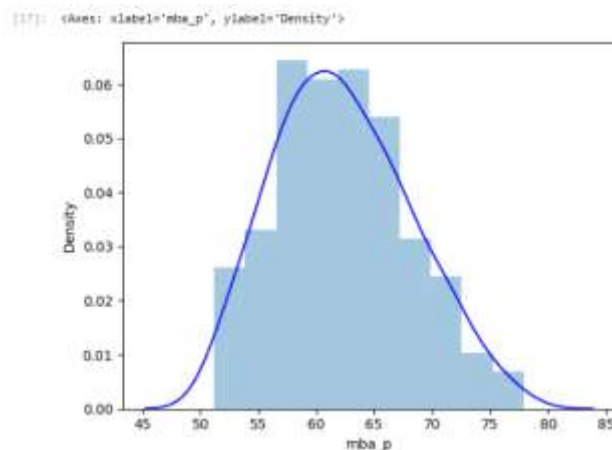
Placement Dataset – Data Visualization Analysis

	sl_no	gender	ssc_p	ssc_b	hsc_p	hsc_b	hsc_s	degree_p	degree_t	workex	etest_p	specialisation	mba_p	status	salary
0	1	M	67.00	Others	91.00	Others	Commerce	58.00	Sci&Tech	No	55.0	Mkt&HR	58.80	Placed	270000.0
1	2	M	79.33	Central	78.33	Others	Science	77.48	Sci&Tech	Yes	86.5	Mkt&Fin	66.28	Placed	200000.0
2	3	M	65.00	Central	68.00	Central	Arts	64.00	Comm&Mgmt	No	75.0	Mkt&Fin	57.80	Placed	250000.0
3	4	M	56.00	Central	52.00	Central	Science	52.00	Sci&Tech	No	66.0	Mkt&HR	59.43	Not Placed	NaN
4	5	M	85.80	Central	73.60	Central	Commerce	73.30	Comm&Mgmt	No	96.8	Mkt&Fin	55.50	Placed	425000.0
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
210	211	M	80.60	Others	82.00	Others	Commerce	77.60	Comm&Mgmt	No	91.0	Mkt&Fin	74.49	Placed	400000.0
211	212	M	58.00	Others	60.00	Others	Science	72.00	Sci&Tech	No	74.0	Mkt&Fin	53.62	Placed	275000.0
212	213	M	67.00	Others	67.00	Others	Commerce	73.00	Comm&Mgmt	Yes	59.0	Mkt&Fin	69.72	Placed	295000.0
213	214	F	74.00	Others	66.00	Others	Commerce	58.00	Comm&Mgmt	No	70.0	Mkt&HR	60.23	Placed	204000.0
214	215	M	62.00	Central	58.00	Others	Science	53.00	Comm&Mgmt	No	89.0	Mkt&HR	60.22	Not Placed	NaN

215 rows × 15 columns

The dataset contains **215 rows and 15 columns**. For the initial analysis, I performed **univariate analysis** using histogram plots to understand the distribution of individual variables.

UNIVARIATE ANALYSIS- HISTOGRAM



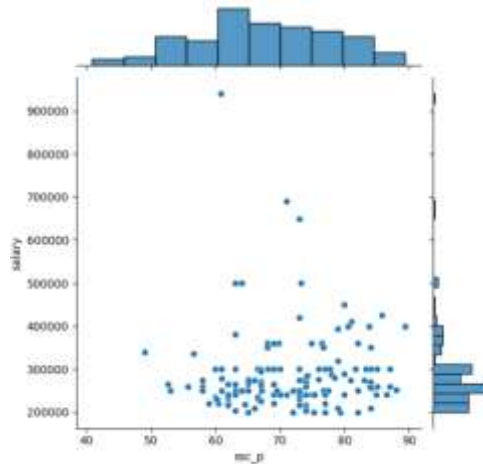
I analyzed the **MBA percentage (mba_p)** using a histogram. The distribution shows that most students scored between **55 and 70 marks**, indicating that the majority of students fall within this range.

The number of students scoring between **45 and 55 marks** is comparatively lower, representing the lower score range. Similarly, the number of students scoring in the **72 to 80 marks** range is also relatively small, indicating fewer high scorers.

Overall, most students have **average scores**, while both the lower and higher score ranges contain fewer students. This pattern suggests that the MBA percentage distribution is **approximately normally distributed**, with most observations concentrated around the middle range.

BIVARIATE ANALYSIS

Joint plot



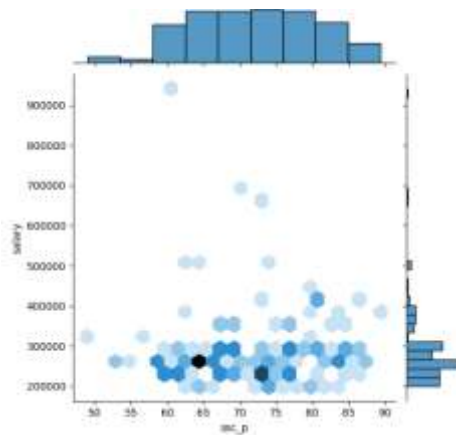
Next, I performed **bivariate analysis** using a **joint plot**. A joint plot is used to analyze the relationship between **two variables** by combining multiple plots in a single visualization. It typically combines a **scatter plot** with **histograms** to show both the relationship between the variables and their individual distributions.

First, I compared **SSLC percentage (ssc_p)** with **salary** using the joint plot.

From the visualization, it can be observed that most students who scored between **62 and 85 in SSLC marks** fall within the **2 to 4 LPA salary range**. A few students who scored between **65 and 75** fall within the **5 to 7 LPA salary range**, and only **one student with around 65 marks** falls in the **9 LPA salary category**.

The **SSLC marks distribution appears approximately normal**, while the **salary distribution is concentrated mostly between 2 to 5 LPA**. Only a very small number of students receive the highest salary package of **9 LPA**.

Overall, the visualization suggests that most students are placed in the **2 to 5 LPA salary range**, regardless of small variations in SSLC scores.

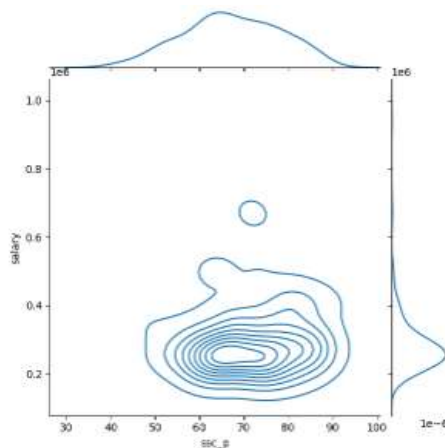


Hex Plot Analysis

Next, I changed the **joint plot kind** to "hex" to visualize the data using a **hexbin plot**.

The hexbin plot represents data density using **hexagonal bins**. Areas with a **higher concentration of data points appear darker**, while areas with fewer observations appear lighter.

From this visualization, we can clearly observe the regions where **most students fall in terms of SSLC marks and salary range**, with darker hexagons indicating higher student concentration.

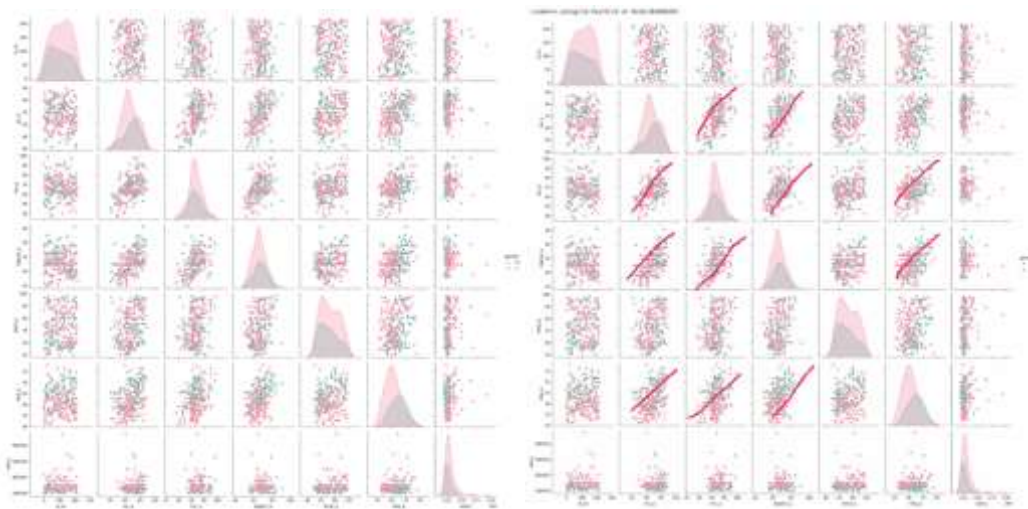


KDE (Kernel Density Estimation) Analysis

Next, I changed the **joint plot kind** to **KDE (Kernel Density Estimation)**.

In this visualization, the relationship between SSLC marks and salary is represented using **density curves**. KDE plots estimate the probability density of the data and help visualize where observations are more concentrated.

Pair plot



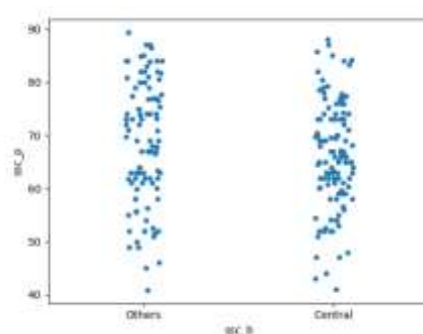
examined the entire dataset using pair plot, including gender distribution and variance. I observed that the academic score features — SSC, HSC, Degree, and MBA percentages — show a linear relationship with each other.

This indicates that multiple features are correlated, which suggests the presence of **multicollinearity** in the dataset. Since multicollinearity can negatively affect model performance, it needs to be handled before model training.

To identify the severity of multicollinearity, we can use the **Variance Inflation Factor (VIF)**. Based on the VIF values, features with high multicollinearity can be addressed using techniques such as removing highly correlated variables or applying dimensionality reduction methods.

Categorical Plots

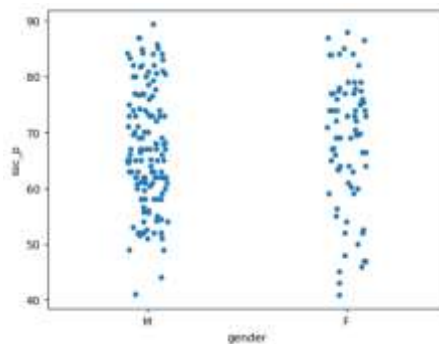
Strip plot



To analyze categorical variables, I used visualization techniques such as **strip plots** and **swarm plots**.

In the first analysis, I used **SSLC Board** as the categorical variable on the x-axis and **SSLC Pass Marks** as the numerical variable on the y-axis. From the visualization, I observed that most students from the **Central Board** scored between **50 and 85 marks**. Students from **other boards** mostly scored between **60 and 90 marks**.

Additionally, only a few students from both boards scored below **50 marks**. Overall, most students performed well, but the **highest scorers were mainly from the other boards**.

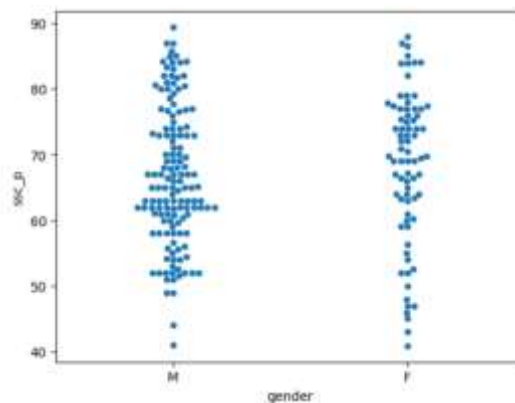


In the second analysis, I compared **Gender and SSCL Pass Marks** using a strip plot. The graph shows that the **number of male students is higher than female students** in the dataset.

Most **male students** scored **between 50 and 85 marks**, with only a few scoring in the **40–50 range**, and one student scoring around **89 marks**.

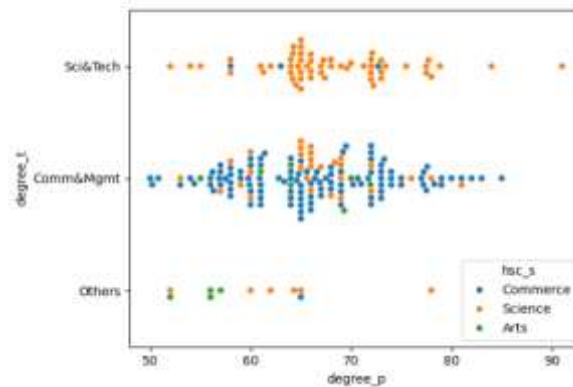
For **female students**, most scores fall between **60 and 80 marks**, while the **top performers** scored **between 80 and 90 marks**. A few students scored between **40 and 50 marks**.

Swarm plot



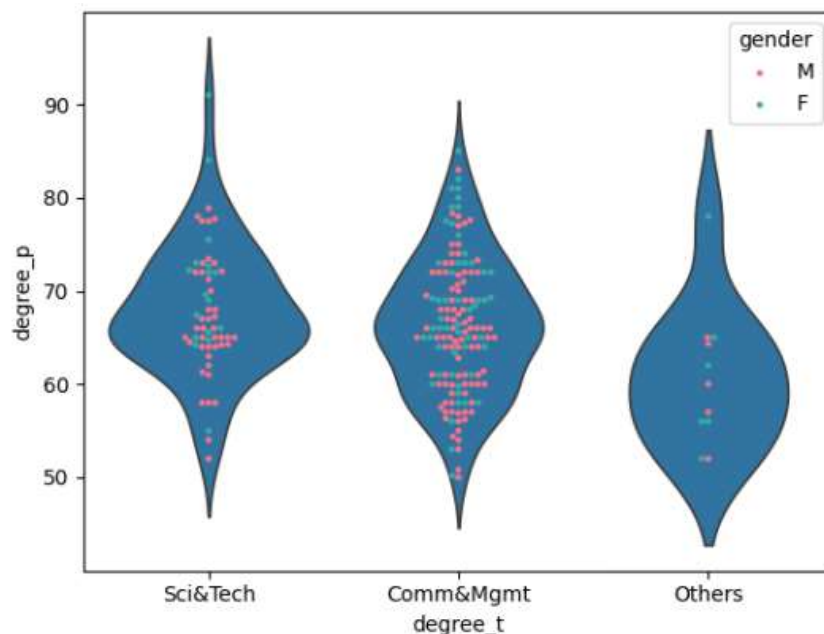
A **swarm plot** is similar to a scatter plot but avoids overlapping points, providing a clear view of data distribution. It is particularly useful for visualizing **numerical values against categorical variables**.

I first analyzed **Gender vs SSCL Pass Marks** using a swarm plot. The analysis revealed that male students were more numerous than female students. Most male students scored between **50 and 85 marks**, with one top performer at **90**, while only two scored in the **40–50 range**. Female students mostly scored **60–80**, with a few scoring below **50**.



Next, I compared **Degree Specialization vs Degree Pass Marks**, considering students' **HSC streams** (Commerce, Science, Arts). Students choosing **Science & Technology degrees** mostly came from Science or Commerce HSC streams. Most **Commerce & Management degree students** came from Commerce HSC, with some from Science and Arts. Few students chose **Other specializations**, mainly Science HSC students.

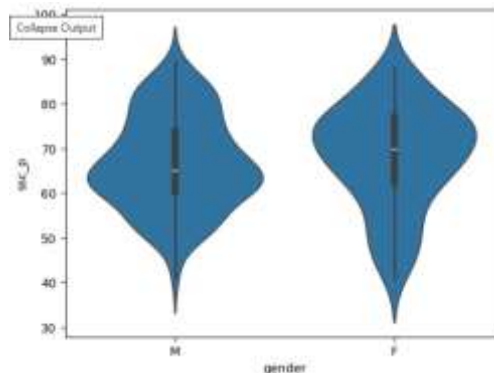
Performance-wise, most students in **Commerce & Management** scored **55–85**, while in **Science & Technology**, scores were concentrated between **65–80**, with only two top performers above **85**. **Other specializations** ranged from **50–80**.



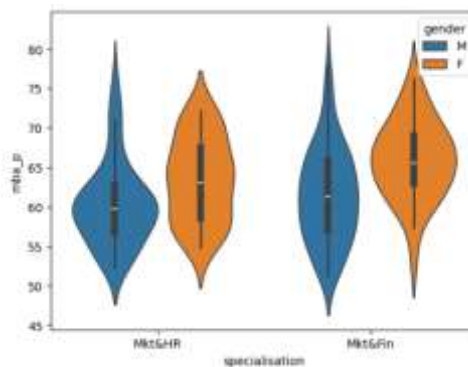
Combining swarm plots with **violin or box plots** provided richer insights. From the violin plot, it was clear that **top performers were mostly from Science & Technology**, while **Commerce & Management** was the most popular choice. Male students were more represented in Commerce & Management, while gender distribution was balanced in Science & Technology and Other specializations.

Distribution Observation

Violin Plot



I analyzed **SSLC Pass Marks vs Gender** using a **violin plot**. For male students, the distribution was **peaked between 55–75**, with few scoring below **50** or above **75**. For female students, most scored **65–85**, with top performers scoring **85–95**, and only a few below **60**. This indicates that **female students performed better overall in SSLC**.

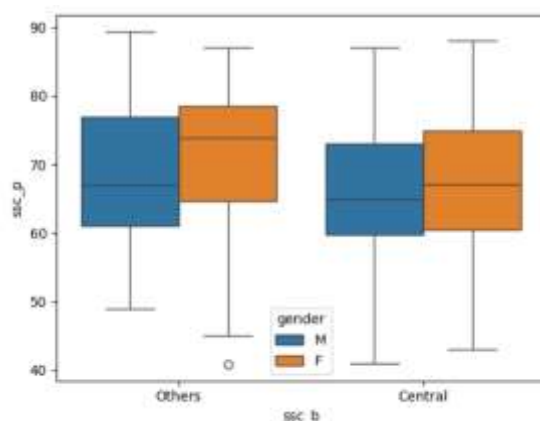


Next, I analyzed **MBA Pass Marks vs Specialization**, differentiating by **gender**. The specializations considered were **Marketing & HR** and **Marketing & Finance**.

- In **Marketing & HR**, most male students scored **55–65**, and most female students scored **55–70**. Top performers were males (**75–80**) while top females scored **75**. Low performers scored below **50** (male) and **50** (female).
- In **Marketing & Finance**, males scored **55–67**, females **62–73**, with top performers males **>80** and females **80**, and low performers similar to the previous specialization.

Overall, **female students performed better on average** across both specializations, but **top performers were male students**. Additionally, students in **Marketing & Finance performed better** than in Marketing & HR.

Box Plot



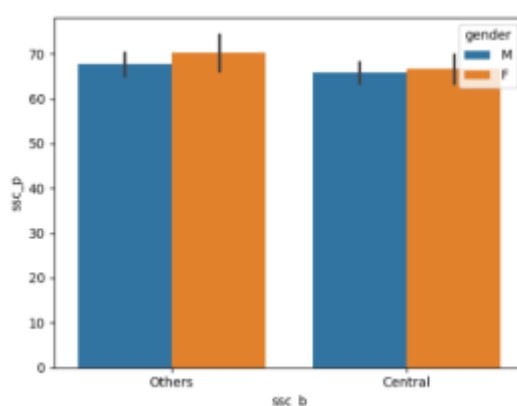
Box Plot Analysis of SSLC Marks by Board and Gender

By analysing the dataset using a **box plot visualization**, we compare SSLC pass marks across different boards and genders.

For students from **other boards**, the results show that male students have a highest score of **89**, while female students have a highest score of **85**. Most male students scored between **62 and 75**, with an average of **around 67**. Female students mostly scored between **65 and 78**, with an average of **around 75**, one female student low scored around 40 (**lower range outlier**) and all female students scored **above 45**, all male students scored **above 49**.

Moving on to students who chose the **Central board**, all male students scored **above 40**, and all female students scored **above 42**. The highest score among male students is **86**, while the highest score among female students is **87**. Most male students scored between **62 and 73**, with an average of **around 65**. Female students mostly scored between **62 and 75**, with an average of **around 67**.

Bar Plot



Bar Plot Analysis of SSLC Pass Marks by Board and Gender

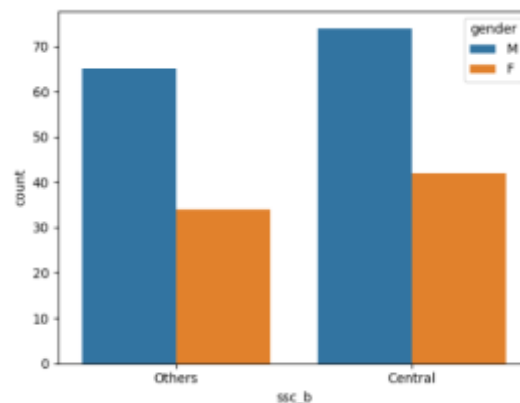
Using a **bar plot**, we analyzed SSLC pass marks with respect to gender across different boards.

From the visualization, it is observed that **female students performed better in the “Other” boards**, achieving higher SSLC pass marks compared to male students.

Similarly, in the **Central board**, female students also performed slightly better than male students.

Overall, when comparing both boards, **students from other boards performed better than those from the Central board** based on the SSLC pass marks.

Count plot



Count Plot Analysis of SSLC Board by Gender

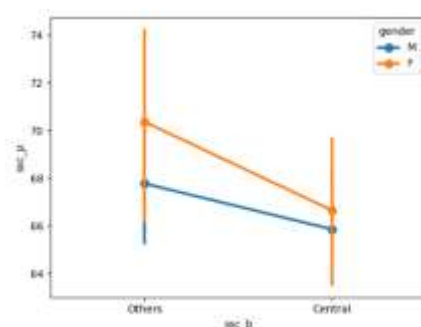
Using a **count plot**, we analyzed the frequency of students choosing different SSLC boards by gender.

For students who chose the **Other boards**, the number of male students is **65**, while the number of female students is **35**.

For students who chose the **Central board**, the number of male students is **above 70**, and the number of female students is **40**.

From this visualization, we can observe that **male students are more represented than female students in both boards**.

Point Plot



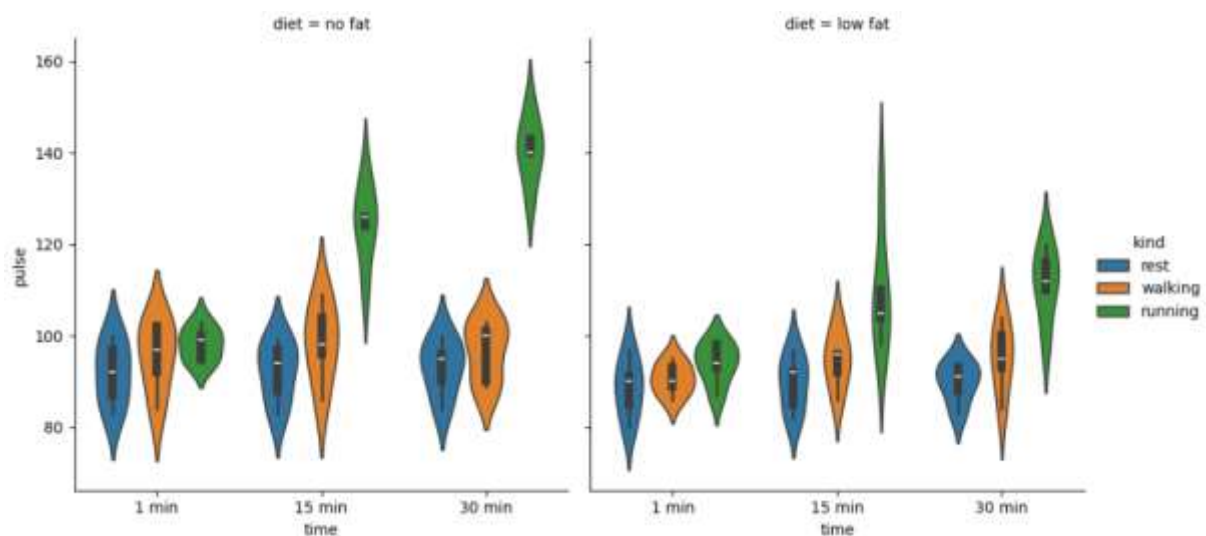
Point Plot Analysis of SSLC Board and SSLC Pass Marks

I analysed the relationship between the **SSLC board** and **SSLC pass marks** using a **point plot**. A point plot is used to visualize the **central tendency of a numerical variable across different levels of one or more categorical variables**. The estimator used in a point plot is not limited to the **mean**. We can also calculate other statistical measures such as the **median, mode, or standard deviation** by

specifying them in the **estimator parameter**. Additionally, several other parameters can be used to extract deeper insights from the visualization.

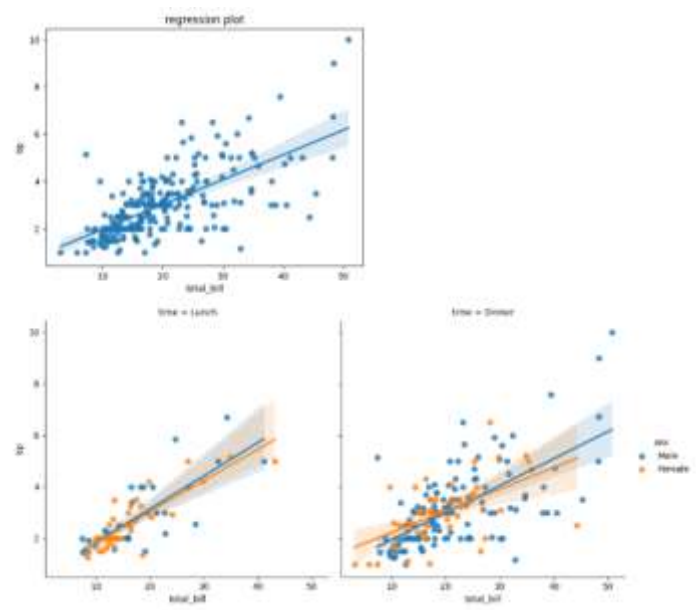
In this analysis, I compared Central and Other SSLC boards with respect to SSLC pass marks, while also differentiating the data by gender (male and female). From the graph, it can be observed that In Other SSLC boards, female students have an average score of around 70 marks. male students have an average score of around 68 marks. In Central SSLC boards, female students have an average score of around 67 marks. male students have an average score of around 66 marks.

Cat plot



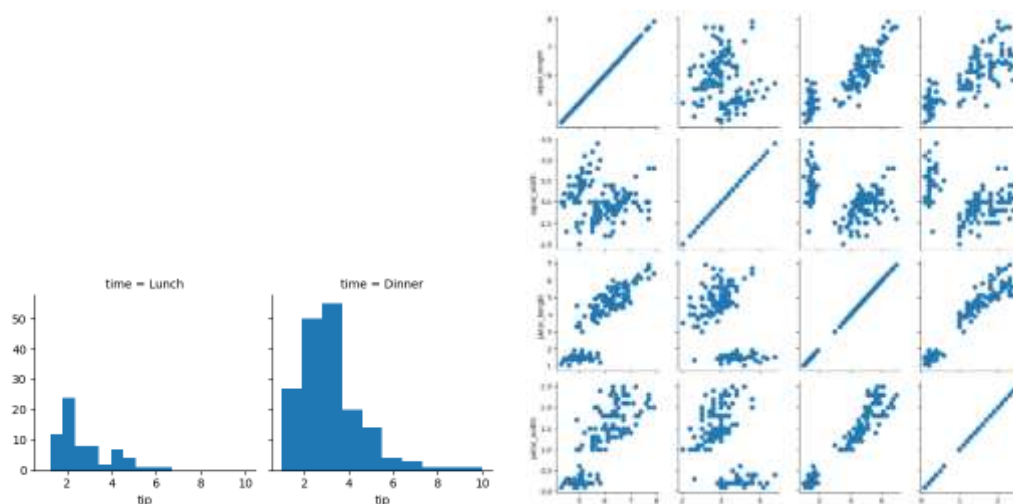
I analyzed the exercise dataset using a **Seaborn catplot with a violin plot**, comparing **time versus pulse rate**, while differentiating the type of exercise (rest, walking, and running) and faceting the visualization by **diet type (No-Fat and Low-Fat)**. The violin plot helps visualize the distribution of pulse rates for different exercise durations of **1, 15, and 30 minutes**. For participants following a **No-Fat diet**, the pulse rate during **1 minute** ranged from **85–99 for rest, 90–102 for walking, and 95–100 for running**. At **15 minutes**, the pulse ranged from **86–97 for rest, 97–103 for walking, and 122–125 for running**. At **30 minutes**, the pulse ranged from **90–96 for rest, 92–101 for walking, and 142–145 for running**, showing a significant increase during running with longer durations. For participants following a **Low-Fat diet**, the pulse rate during **1 minute** ranged from **83–90 for rest, 86–96 for walking, and 96–99 for running**. At **15 minutes**, the pulse ranged from **84–90 for rest, 90–97 for walking, and 103–115 for running**. At **30 minutes**, the pulse ranged from **80–95 for rest, 90–103 for walking, and 105–125 for running**. Overall, the analysis shows that **pulse rate increases with both exercise intensity and duration**, especially during running, and participants with a **No-Fat diet** tend to exhibit slightly higher pulse rates during longer running sessions compared to those with a **Low-Fat diet**.

Regplot and Implot



I analyzed the relationship between **total bill and tips using a regression plot**, which displays data points as a scatter plot with a fitted regression line using a simple linear model. I used an **LM plot** to further analyze the data by separating it based on **time (Lunch and Dinner)** and **gender (Male and Female)**. From the visualization, most customers visited during **dinner time**, and comparatively **male customers were more frequent**. Most customers had a **total bill between 10 and 40** and gave **tips between 2 and 6**, while a few male customers during dinner had **higher bills between 40 and 50 with tips ranging from 7 to 10**. Overall, the plot shows a **positive relationship between total bill and tips**, meaning that tips generally increase as the total bill increases.

FacetGrid and PairGrid



A **FacetGrid** is used to plot **data distributions for categorical variables** by mapping different types of plots, such as **histograms, scatter plots, box plots, or bar plots**, to each facet. In contrast, a **PairGrid** is used to explore the **relationships among multiple numerical variables**, allowing us to visualize pairwise relationships across several variables in a single grid of plots.

